



Influenza vaccination induces autoimmunity against orexinergic neurons in a mouse model for narcolepsy

Raphaël Bernard-Valnet,^{1,2,†} David Frieser,^{1,†} Xuan-Hung Nguyen,^{1,3,†} Leila Khajavi,¹ Clémence Quériault,¹ Sébastien Arthaud,⁴ Silvia Melzi,⁴ Maxime Fusade-Boyer,⁵ Frederick Masson,¹ Matthias Zytnecki,⁶ Abdelhadi Saoudi,¹ Yves Dauvilliers,⁷ Christelle Peyron,⁴ Jan Bauer⁸ and Roland S. Liblau^{1,9}

[†]These authors contributed equally to this work.

Narcolepsy with cataplexy or narcolepsy type 1 is a disabling chronic sleep disorder resulting from the destruction of orexinergic neurons in the hypothalamus. The tight association of narcolepsy with HLA-DQB1*06:02 strongly suggest an autoimmune origin to this disease. Furthermore, converging epidemiological studies have identified an increased incidence for narcolepsy in Europe following Pandemrix[®] vaccination against the 2009–2010 pandemic ‘influenza’ virus strain. The potential immunological link between the Pandemrix[®] vaccination and narcolepsy remains, however, unknown. Deciphering these mechanisms may reveal pathways potentially at play in most cases of narcolepsy. Here, we developed a mouse model allowing to track and study the T-cell response against ‘influenza’ virus haemagglutinin, which was selectively expressed in the orexinergic neurons as a new self-antigen. Pandemrix[®] vaccination in this mouse model resulted in hypothalamic inflammation and selective destruction of orexin-producing neurons. Further investigations on the relative contribution of T-cell subsets in this process revealed that haemagglutinin-specific CD4 T cells were necessary for the development of hypothalamic inflammation, but insufficient for killing orexinergic neurons. Conversely, haemagglutinin-specific CD8 T cells could not initiate inflammation but were the effectors of the destruction of orexinergic neurons. Additional studies revealed pathways potentially involved in the disease process. Notably, the interferon- γ pathway was proven essential, as interferon- γ -deficient CD8 T cells were unable to elicit the loss of orexinergic neurons.

Our work demonstrates that an immunopathological process mimicking narcolepsy can be elicited by immune cross-reactivity between a vaccine antigen and a neuronal self-antigen. This process relies on a synergy between autoreactive CD4 and CD8 T cells for disease development. This work furthers our understanding of the mechanisms and pathways potentially involved in the development of a neurological side effect due to a vaccine and, likely, to narcolepsy in general.

- 1 Toulouse Institute for Infectious and Inflammatory Diseases (Infinity), University of Toulouse, CNRS, INSERM, UPS, Toulouse, France
- 2 Service of Neurology, Clinical Neurosciences Department, Lausanne University Hospital and Lausanne University, Lausanne, Switzerland
- 3 Vinmec Institute of Applied Science and Regenerative Medicine, Vinmec Healthcare System, Hanoi, Vietnam
- 4 INSERM U1028, CNRS UMR 5292, Center for Research in Neuroscience, University of Lyon 1, Bron, France
- 5 IHAP, UMR1225, Université de Toulouse, INRAe, ENVT, Toulouse, France
- 6 Unité de Mathématiques et Informatique Appliquées, INRAE, Castanet-Tolosan, France
- 7 National Reference Center for Orphan Diseases, Narcolepsy, Idiopathic hypersomnia and Kleine-Levin Syndrome, Department of Neurology, Gui-de-Chauliac Hospital, CHU de Montpellier, INSERM U1061, Montpellier, France

Received May 26, 2021. Revised November 03, 2021. Accepted November 24, 2021. Advance access publication May 13, 2022

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8 Center for Brain Research, Medical University of Vienna, Vienna, Austria

9 Department of Immunology, Toulouse University Hospitals, Toulouse, France

Correspondence to: Prof. Roland Liblau

Toulouse Institute for infectious and inflammatory diseases

CHU Purpan BP 3028, 31024 Toulouse cedex 03

Toulouse, France

E-mail: roland.liblau@inserm.fr

Keywords: narcolepsy; orexinergic neurons; autoimmunity; T cells; interferon-gamma

Abbreviations: HA = ‘influenza’ haemagglutinin; HLA = human leukocyte antigen; IFN- γ = interferon-gamma; MCH = melanin concentrating hormone; NT1 = narcolepsy type 1; TCR = T-cell receptor

Introduction

Narcolepsy type 1 (NT1), is a rare, chronic and disabling neurological disease, usually starting at adolescence. It is characterized by excessive daytime sleepiness, cataplexy (sudden loss of muscle tone triggered by emotions) and fragmented nocturnal sleep.^{1,2} Post-mortem pathological studies revealed that NT1 results from a profound and selective loss of a small population of neurons in the lateral hypothalamus characterized by the production of the orexin/hypocretin neuropeptides.^{3,4} To date, there is no curative treatment for NT1, and patients usually require life-long symptomatic therapies.

The aetiology of NT1 is multifactorial, with disease risk determined both by genetics and environmental factors. Although NT1 pathophysiology remains to be deciphered, converging evidence supports an immune-mediated basis. Indeed, the disease is strongly associated worldwide with the human leukocyte antigen (HLA) class II haplotype HLA-DQA1*01:02-HLA-DQB1*06:02 (odds ratio of 250) and weaker associations have been identified with HLA class I alleles.^{5–7} Moreover, NT1 is associated with polymorphisms in other immune-related genes, most notably T-cell receptor (TCR)- α and TCR- β chains, perforin and cathepsin H.^{5–10} The nature of the genes involved in disease susceptibility suggests the importance of both CD4 and CD8 T cells in the disease process. Additionally, an increased incidence of NT1 in children and adolescents was associated with the 2009–2010 H1N1 ‘influenza’ vaccination campaign using Pandemrix[®] in several European countries.^{11–14} A reassessment of the risk of narcolepsy in children in England after receipt of the AS03-adjuvanted H1N1 pandemic vaccine concluded to a 7-fold elevated risk of onset of narcolepsy after Pandemrix[®], which was confined to the first year after vaccination.¹⁴ Interestingly, a recent report suggests that treatment with immune checkpoint inhibitors could cause narcolepsy in genetically predisposed persons.¹⁵ Thus, antigen-specific as well as more general immune stimulation have been associated with development of narcolepsy.

The striking association with HLA and with a specific vaccine, together with the selective nature of the neuronal destruction observed in NT1 strongly supports an autoimmune basis. So far, efforts to identify pathogenic autoantibodies have not been successful. Recent data, however, support the involvement of autoreactive T cells in NT1 pathogenesis. Our mouse model of immune-mediated NT1 provided clear proof-of-principle that a CD8 T cell-mediated autoimmune process could lead to an NT1-like phenotype, associating orexinergic neuron death, sleep attacks and cataplexy episodes.¹⁶ Accordingly, NT1 patients showed a higher frequency of CD8 T cells recognizing NT1-related candidate

proteins, identified using HLA:peptide multimers, when compared to HLA-DQ-matched controls.¹⁷ Moreover, orexin-specific CD8 T-cell clones were detected in the CSF of one patient.¹⁸ In addition, increased frequencies of CD4 T cells of the Th1 phenotype recognizing orexin peptides have been identified in the blood of patients with NT1 compared to HLA-DQB1*06:02-matched healthy controls.^{18,19} Single-cell TCR sequencing suggested clonal expansion and molecular mimicry between orexin and ‘influenza’ virus peptides.¹⁹ Overall, these findings collectively point to the involvement of both CD4 and CD8 in the pathogenesis of NT1, in line with the increased propensity of blood CD4 and CD8 T cells to produce B cell-supporting cytokines in NT1 patients as compared to matched controls.²⁰ However, the interplay between CD4 and CD8 T cells that elicits hypothalamic tissue damage is completely unknown. In addition, it remains unclear whether and how potentially pathogenic autoreactive T cells can be activated by exposure to the Pandemrix[®] vaccine.

In this study, we leveraged our mouse model expressing ‘influenza’ haemagglutinin (HA) selectively in orexinergic hypothalamic neurons and TCR-transgenic mouse lines in which a large proportion of either CD4 or CD8 T cells recognize HA, to identify whether the Pandemrix[®] vaccine could elicit hypothalamic inflammation and loss of orexin-producing neurons and tease apart the mechanisms involved. Using histology, cellular immunology, transcriptomic of sorted T-cell populations, and loss-of-function experiments we investigated the potential contribution of CD4 and CD8 T-cell subsets in the pathogenic processes. The data indicate that autoreactive CD4 T cells are essential to promote hypothalamic inflammation but do not, by themselves, induce loss of orexin-producing neurons. Conversely, autoreactive cytotoxic CD8 T cells are necessary for the selective destruction of orexin-producing neurons.

Materials and methods

Mouse strains

The (BALB/cJ x C57Bl/6J)F1 Orex-HA mice are a double transgenic strain expressing HA of an H1N1 influenza virus (A/Puerto Rico/8/34/Mount Sinai) mainly in Orexin neurons¹⁶. The TCR-transgenic mice (BALB/cJ x C57Bl/6J)F1 6.5-TCR and CL4-TCR, expressing an I-E^d-restricted HA_{110–119}-specific TCR and a K^d-restricted HA_{512–520}-specific TCR, respectively, have been previously described.^{21,22} The TCR-transgenic mice harbor the CD45.1 allotype, whereas the recipient mice only express CD45.2. All mice were kept in pathogen-free conditions

within the UMS006 animal facility, Toulouse, France. All animal experiments were performed in accordance with the European Union guidelines following approval of the local ethics committee (APAFIS#770-2017010511127939V4).

Naïve T-cell transfer and mouse vaccination

Naïve HA-specific CD4 and CD8 T cells were purified from 6.5-TCR and CL4-TCR mice, respectively. Spleens and lymph nodes were collected, and the relevant T-cell subset was purified (Dynabeads Untouched Mouse CD4 or CD8 Cells kit; Invitrogen). CD4 T cells were then incubated with anti-CD25 (clone PC61) monoclonal antibody (mAb) for regulatory T-cell depletion. Naïve CD4 or CD8 T cells were further purified based on expression of CD62L (Miltenyi Biotech). Naïve CD4 and/or CD8 T cells (5×10^6 each) were injected intravenously into the recipient mice. For *in vivo* proliferation experiments, T cells were stained before transfer with either CFSE or CellTrace Violet (Invitrogen).

Twenty-four hours after transfer of HA-specific T cells, Orex-HA mice and their littermate counterparts were injected in the gluteal muscle with 50 μ l of either PBS, AS03 adjuvant or commercial Pandemrix® (kindly provided by GlaxoSmith Kline) containing split virion, inactivated ‘influenza’ A/California/07/2009 (H1N1).

In some experiments, depleting anti-CD8 mAbs (YTS 169.4; 200 μ g/injection) or control rat IgG2b from BioXCell InvivoMab were injected every 5 days from D5 onwards until the end of the experiment.

Isolation of hypothalamus-infiltrating mononuclear cells

Mice were deeply anaesthetized with ketamine (100 mg/kg) and xylazine (10 mg/kg) and perfused intracardially with PBS. Brains were immediately removed, and a hypothalamic block was dissected out, homogenized, and digested with collagenase D (1 mg/ml) and DNase I (10 mg/ml). Hypothalamus-infiltrating mononuclear cells were then collected using Percoll density separation.

Fluorescence activated cell sorting analysis

Cells were harvested from hypothalamus, spleen, draining lymph nodes (i.e. inguinal and sciatic lymph nodes) and cervical lymph nodes at different time points after vaccination. Prior to staining, cells were incubated with anti-CD16 (Fc γ R-Blocker; 4GG2: in-house production). Cells were then stained for cell surface markers and, in some experiments, intracellular molecules. Intracellular cytokine staining was performed following 4 h stimulation with PMA (50 ng/ml, Sigma-Aldrich) and ionomycin (50 ng/ml, Sigma-Aldrich) in the presence of Golgi-plug (0.4% v/v, BD Biosciences). Cell suspensions were stained by using a viability dye and anti-CD11b (M1/70, eBioscience), anti-Thy1.2 (53-2.1, eBioscience), anti-MHC-class II (M5/114.15.2, eBioscience), anti-FoxP3 (FJK-16s, eBioscience), anti-granzyme B (NGZB, eBioscience), anti-CD4 (RM4-5, BD Biosciences), anti-CD8 α (53-6.7, BD Biosciences), anti-CD45.2 (104, BD Biosciences), anti-CD45.1 (A20, BD Biosciences), anti-CD44 (IM7, BD Biosciences), anti-CD62L (MEL14, BD Biosciences), anti-CD11c (N418, BioLegend), anti-Ly6C (HK1.4, BioLegend), anti-CD8 α (YTS156.7.7, BioLegend), anti-IL17A (TC11-18H10, BD Biosciences), anti-IFN γ (XMG1.2, BD Biosciences), anti-GM-CSF (MP1-22E9, BD Biosciences), anti-TNF α (MP6-XT22, BD Biosciences) monoclonal antibodies. Data were collected on LSRII Fortessa (Becton Dickinson) and analysed with FlowJo software (Tree Star). The t-distributed stochastic neighbour embedding plots were done using CytoBank (CytoBank Inc.).

Peptide-restimulation of HA-specific T cells in hypothalamic cell suspensions.

Mice were anaesthetized and perfused intracardially with PBS. Brains were removed and a hypothalamic block was dissected out, homogenized and digested in HBSS medium with collagenase D (1 mg/ml), TLCK (30 μ g/ml) and DNase I (20 μ g/ml), all from Roche® Diagnostics. After digestion the hypothalamic cell suspension was resuspended in Percoll (30%; GE Healthcare) and spun at a centrifugation step of 1590g for 30 min without break. Myelin and adipose debris were aspirated, and the cells collected. Hypothalamic immune cell suspensions were stimulated with HA_{512–520}-peptide and HA_{110–119}-peptide (Genecust; 100 μ g/ml each) in the presence of Golgi-stop (0.07% v/v, BD Biosciences).

Haemagglutination inhibition assay

Mouse sera from vaccinated and PBS- or AS03-injected control mice were tested for anti-‘influenza’ antibodies using two viral strains: A/Puerto Rico/8/1934(H1N1) (PR8) and A/Michigan/45/2015(H1N1), kind gift of Dr Richard Webby. Haemagglutination inhibition assays (HAI) were performed as previously described²³ using 1% chicken red blood cells. The HAI titres were determined as the reciprocal dilution of the last well that contained non-agglutinated red blood cells.

Histopathology

Tissues were fixed in 4% paraformaldehyde, conserved in 70% ethanol, and embedded in paraffin. Immunohistochemical and Immunofluorescence staining of 5- μ m thick coronal sections was performed as described elsewhere²⁴ using the following primary antibodies: rabbit anti-orexin A (Phoenix Pharmaceuticals, #H-003-30), rabbit anti-CD3 (SP7; Neomarkers), rabbit anti-CD8 α (Abcam, ab209775), rabbit anti-Iba-1 (Wako, #019-19741), rabbit anti-phospho-Stat1 (Cell Signaling, #9167) and anti-granzyme B (ab4059, AbCam). All secondary antibodies were from Jackson ImmunoResearch Laboratories. Fluorescent preparations afterwards were stained with 4',6'-diamidino-2-phenylindole (DAPI, Sigma), embedded and examined using a confocal laser scan microscope (Leica SP5, Leica) equipped with lasers for 504, 488, 543 and 633 nm excitation. Scanning for DAPI (504 nm), Cy2 (488 nm), Cy3 (543 nm) and Cy5 (660 nm) was performed sequentially to rule out fluorescence bleed-through. For β 2-microglobulin, or-oxin, CD8 co-staining, orexin A staining was first performed on untreated sections followed by biotinylated donkey-anti-rabbit secondary antibody. This was followed by consecutive incubation with avidin-peroxidase and biotinylated tyramide. Next the sections were pre-treated in EDTA 8.5 for 1 h in a household cooking device (Braun). Afterwards, rabbit-anti-CD8 α and goat-anti- β 2-microglobulin (M-20, Santa Cruz) were incubated for 1 h at RT. The staining was completed by shred incubation with Cy3-conjugated donkey-anti-rabbit, Cy2-conjugated donkey-anti-goat and Cy5-conjugated streptavidin. Image acquisition was performed with 3DHistec Panoramic 250 slide scanner (3DHistec Ltd.).

For neuronal enumeration, brains were fixed in 4% paraformaldehyde and 30- μ m thick coronal free-floating sections performed at cryostat. Immunohistochemical detection of orexin+ and melanin-concentrating hormone (MCH)+ neurons was performed by using the antibodies described above. Neuron enumeration was performed on 1/12th of the entire hypothalamus (seven sections of 30 μ m thickness per brain) by using Mercator software (Explora Nova). Alternatively, 10 hypothalamic sections (5 μ m) per animal

were stained for orexin by immunohistochemistry. Image acquisition was performed with 3DHistec Panoramic 250 slide scanner (3DHistec Ltd.). Enumeration was performed using the automatic count tool with CaseViewer software (3DHistec Ltd.).

RNA sequencing of T-cell populations

Orex-HA mice were immunized intramuscularly with Pandemrix[®] vaccine 1 day after the adoptive transfer of naïve (CD45.1⁺CD62L⁺CD25[−]) HA-specific CD4⁺ and CD8⁺ T cells. At Day 14 post-vaccination, endogenous (CD45.1[−]) and transferred (CD45.1⁺) populations of CD44^{high}CD62L^{low}CD4⁺ and CD44^{high}CD62L^{low}CD8⁺ T cells were isolated from spleens, cervical lymph nodes and hypothalami by fluorescence activated cell sorting in four independent experiments involving 20 to 30 mice each.

Total RNA was extracted from the sorted cells using RNeasy Micro Kit (Qiagen). RNA quality and quantity was checked by Eukaryote Total RNA Pico assay (Agilent) and validated with a RIN > 7. Library preparation and RNA-sequencing were performed by the Genomics Facility Basel in Zurich using an Illumina HiSeq 2500, generating 75-bp single-end reads.

Sequences were quality checked using FastQC (version V0.11.2) and FastqScreen (version V0.11.4) prior to aligning to the *Mus musculus* primary genome sequence (Gencode: GRCm38.p5, release M16) using STAR (version V2.4.0i). FastQC, featureCounts (V1.4.5-p1) and PicardTools (version V2.17.3) were used to assess the mapping quality. RSEM (V1.3.0) was used to generate the expression matrix that was then analyzed with DESeq2 (R version V3.4.4, BioConductor version V3.6) to identify the differentially expressed genes (DEGs). ClusterProfiler (R package) was used to classify the rendered DEGs into KEGG pathways, gene ontology (GO) terms, and gene set enrichment analysis (GSEA). These classifications were further interrogated to yield a gene list of interest based on immunological relevance.

Raw fastQ reads were used to extract TCR α and β sequences with MiTCR and to generate chord diagrams.²⁵

Polysomnographic analyses

These combined EEG and EMG recordings were performed as described.¹⁶

Data availability

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

Results

In vivo activation of HA-specific T cells in Pandemrix[®]-vaccinated mice

Our general objective was to investigate whether the Pandemrix[®] vaccine could elicit features of NT1 through molecular mimicry and to decipher the underlying immune pathways. To this end, we made use of our original mouse model, the so-called Orex-HA mice, in which the H1N1 ‘influenza’ virus HA is expressed specifically in orexinergic neurons.¹⁶ To trace the HA-specific T cells following Pandemrix[®] vaccination, the T-cell repertoire of vaccinated mice was spiked with genetically marked (CD45.1⁺) naïve CD4 and CD8 T cells specific for HA, originating from 6.5-TCR and CL4-TCR transgenic mice, respectively (Supplementary Fig. 1A). As a result, on average $0.87 \pm 0.1\%$ of CD4 T cells and $0.75 \pm 0.05\%$ of CD8 T cells in the spleen

of Orex-HA mice were of transferred origin (Supplementary Fig. 1B). Importantly, the I-E^d-restricted 6.5-TCR and the K^d-restricted CL4-TCR recognize HA peptides that are fully shared by the HA subtypes used in Pandemrix[®] and as a transgene in Orex-HA mice (Supplementary Fig. 1B).

To assess whether the human Pandemrix[®] vaccine could activate and expand *in vivo* the naïve CD45.1⁺ HA-specific CD4 and CD8 T cells, T cells were stained with a proliferation dye prior to their transfer in either Orex-HA mice or their littermate controls not expressing HA. Recipient mice were immunized intramuscularly the next day with the Pandemrix[®] vaccine, its AS03 adjuvant or saline. Extensive proliferation of both HA-specific CD4 and CD8 T cells was detected following Pandemrix[®] vaccination, but neither with PBS nor with the adjuvant alone, in the draining lymph nodes (Fig. 1A and B) and spleen (data not shown) of both Orex-HA mice (Fig. 1A and B) and control animals not expressing HA (Supplementary Fig. 2A). The *in vivo* Pandemrix[®]-primed HA-specific T cells acquired an activated phenotype, as illustrated by up-regulation of CD44 and loss of CD62L expression (Fig. 1C and D), and effector functions (Fig. 1E and F). Indeed, the transferred HA-specific CD8 T cells differentiated in IFN γ - and TNF α -producing effectors, whereas transferred HA-specific CD4 T cells acquired either a Th1 or Th17 phenotype (Fig. 1E and F). Similar results were observed following vaccination of control animals not expressing HA (Supplementary Fig. 2B). By contrast, when used alone, AS03 had a modest effect, if any, on the activation and differentiation of HA-specific T cells in both types of recipient animals (Fig. 1B, D, F and Supplementary Fig. 2A and B). Of note, all tested Pandemrix[®]-vaccinated Orex-HA mice ($n=14$) had anti-HA antibody titres ≥ 5120 UI/ml using a pandemic-like H1N1 virus strain, whereas levels in mice injected with the AS03 adjuvant or PBS ($n=6$) were <10 UI/ml. Interestingly, detection of anti-HA antibodies was fully negative in serum of 14 Pandemrix[®]-vaccinated Orex-HA mice when using the PR8/34 H1N1 strain, suggesting lack of cross-reactivity with the HA expressed in neurons.

HA-specific T cells induce selective destruction of orexin⁺ neurons in vaccinated Orex-HA mice

We next assessed, by immunohistochemistry, whether the vaccine-induced activation of HA-specific CD4 and CD8 T cells resulted in CNS inflammation, as a consequence of HA expression in orexinergic neurons. Two weeks after CD4 and CD8 T-cell transfer and Pandemrix[®] vaccination, T-cell infiltration was observed in the hypothalamus in all Orex-HA mice and in none of the non-transgenic controls (Fig. 2A and B). Importantly, T-cell infiltration was minimal or absent in the other investigated brain regions, such as thalamus, hippocampus and cortex (Supplementary Fig. 3A and B).

This finding led us to further characterize, using flow cytometry, the immune cell types accumulating in the CNS of Orex-HA mice. Contrasting with control mice treated in parallel, the CNS of Orex-HA mice harboured large numbers of CD4 and CD8 T cells 2 weeks after vaccination (Fig. 2C and D). The infiltrated T-cell population was dominantly composed of CD8 T cells [mean and range; 54% (35–69)], including both transferred [16% (4–41)] and endogenous CD8 T cells [38% (18–65)]. Transferred CD4 T cells were rare [5% (1–15)] while endogenous CD4 T cells represented 45% (30–64) of CNS-infiltrating T cells (Fig. 2D). Because infiltrated T cells preferentially localized in the hypothalamus of vaccinated Orex-HA mice, we evaluated the kinetics and composition of the T-cell infiltration selectively on dissected hypothalamus. The hypothalamic T-cell infiltration was detected as soon as 7 days post-vaccination, peaked

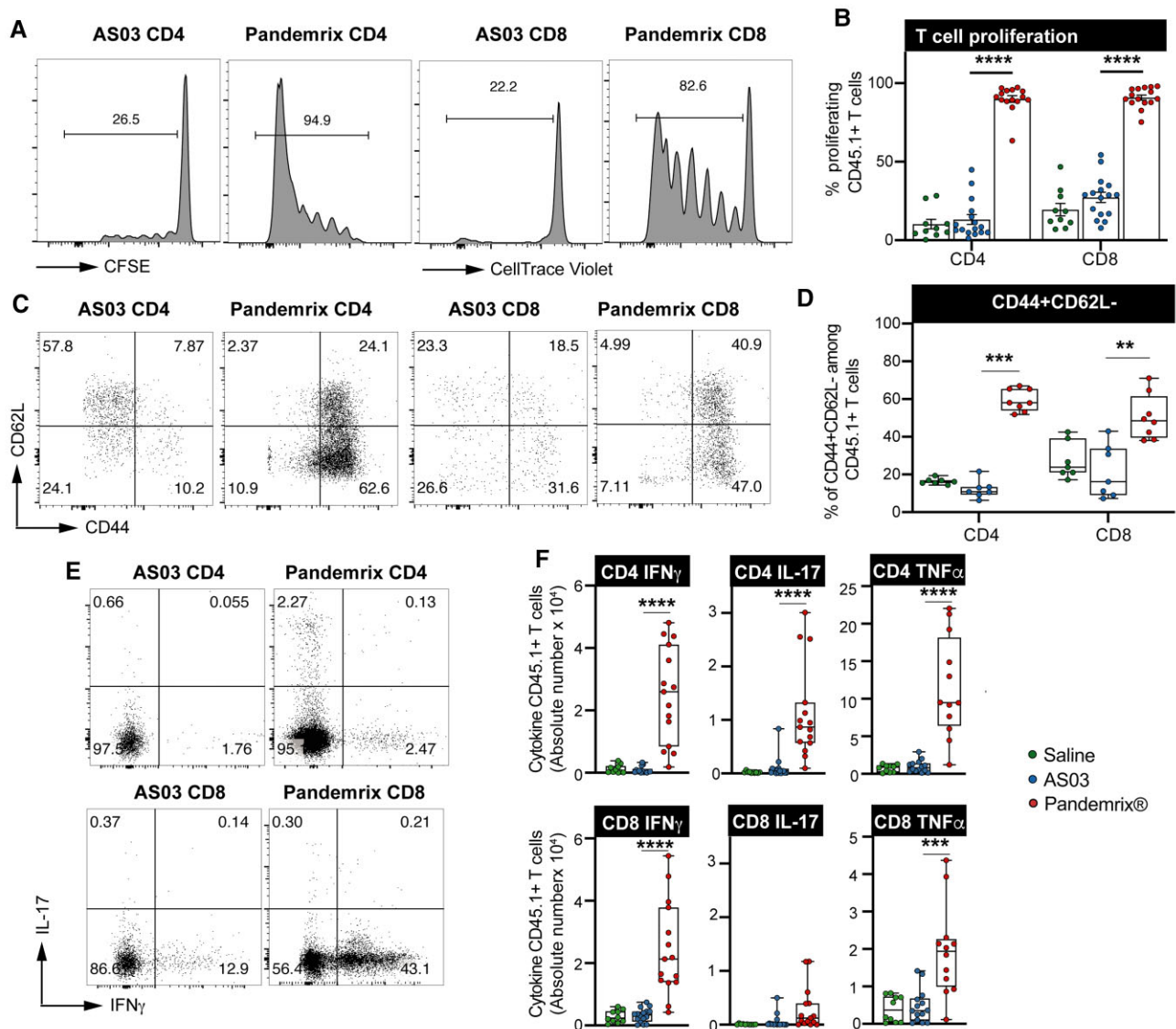


Figure 1 Activation of HA-specific T cells in the draining lymph nodes upon Pandemrix® immunization of Orex-HA mice. (A and B) Evaluation of the *in vivo* proliferation of transferred (CD45.1+) naïve HA-specific CD4 and CD8 T cells of Orex-HA recipient animals 5 days after intramuscular immunization with Pandemrix® (red circles), the AS03 adjuvant (blue circles) or saline (green circles). (A) Representative fluorescence activated cell sorting (FACS) plot of CFSE or Celltrace Violet dilution in CD45.1+ CD4 or CD8 T cells, respectively, and (B) histograms. Results are expressed as mean $\pm$ SEM of 10 to 16 mice per group from six independent experiments. (C) Representative FACS plots of CD44 and CD62L expression on CD45.1+ HA-specific CD4 and CD8 T cells 5 days after intramuscular immunization with Pandemrix® or the AS03 adjuvant. (D) Frequencies of CD45.1+ HA-specific CD4 and CD8 T cells harbouring the CD44+CD62L- activated phenotype. Box plots depict interquartile range and median, and whiskers extend between minimum and maximum values of seven to eight mice per group from two independent experiments. (E) Representative FACS plots of IL-17 and IFN γ expression in CD45.1+ HA-specific CD4 and CD8 T cells 5 days after i.m. immunization with Pandemrix® or the AS03 adjuvant. (F) Absolute numbers of CD45.1+ HA-specific CD4 (top) and CD8 (bottom) T cells producing IFN γ , IL-17, TNF α 5 days after injection of either PBS, AS03 or Pandemrix®. Box plots depict interquartile range and median, and whiskers extend between minimum and maximum values of 10 to 16 mice per group from six independent experiments. Statistical analyses were performed using the Mann-Whitney test. ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

at Day 14 (on average 1562 ± 467 T cells per hypothalamus) and persisted for at least 4 weeks (Fig. 2E). From Day 14 onward, CD8 T cells of transferred or endogenous origin dominated the hypothalamic infiltrates (Fig. 2E). The hypothalamus of control mice at Day 14 contained on average 104 T cells. In addition, a large population of CD11b⁺ MHC class II-expressing myeloid cells was also present in the CNS of vaccinated Orex-HA mice, composed of CD11b⁺ CD45^{dim} microglia and CD11b⁺ CD45^{hi} blood-borne macrophages and dendritic cells (Fig. 2C, F, G and H). Therefore, in Orex-HA mice but not in littermate controls not expressing HA, the

Pandemrix® vaccine elicited a focal and protracted hypothalamic inflammation.

Interestingly, hypothalamus-infiltrating T cells were localized in the vicinity of orexinergic neurons in Orex-HA mice (Fig. 3A). In this inflammatory context, MHC class I molecules were detected at the plasma membrane and intracellularly in orexin⁺ neurons (Fig. 3B). The tight apposition of T cells to orexinergic neurons (Fig. 3A), the expression of activated caspase 3 (Fig. 3C) and the detection of apoptotic nuclei (Fig. 3D and E) in some of the orexinergic neurons led us to investigate whether a loss of orexin-producing

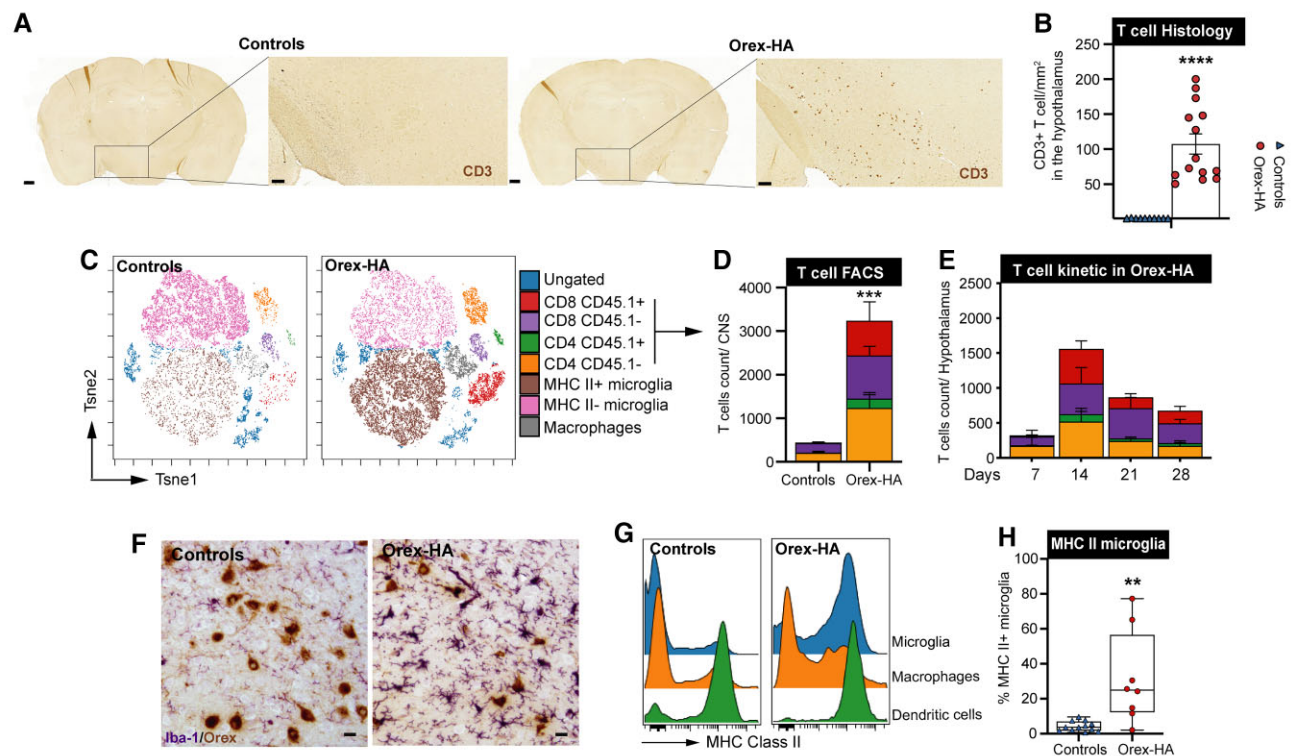


Figure 2 Pandemrix® vaccination following naïve CD4 and CD8 T-cell transfer elicits hypothalamic inflammation in Orex-HA mice. (A) Immunohistochemistry staining for CD3 (brown) in control and Orex-HA mice with a focus on the hypothalamus. Scale bar = 500 µm and 100 µm, respectively. (B) CD3+ T-cell density in the hypothalamus of control (blue triangles) and Orex-HA (red circles) mice 14 days after transfer of both HA-specific CD4 and CD8 T cells and vaccination with Pandemrix®. Results are expressed as mean ± SEM of 10 to 14 mice per group from nine independent experiments. (C) Dimensionality reduction using the t-distributed stochastic neighbour embedding algorithm allows depicting immune subpopulations analysed by flow cytometry in the whole brain of control and Orex-HA mice, 2 weeks after transfer of HA-specific CD4 and CD8 T cells and Pandemrix® immunization. Represented cell populations result from manual gating based on the listed markers. (D) Quantification of the number of endogenous and transgenic T-cell infiltration within the whole brain of control and Orex-HA mice 14 days after Pandemrix® immunization. Results are expressed as mean ± SEM of seven to eight mice per group from two independent experiments. (E) Numbers of infiltrating endogenous and transgenic T cells within dissected hypothalami of Orex-HA mice over time after Pandemrix® immunization. Results are expressed as mean ± SEM of two to four pools of hypothalami for a total of seven to nine mice per time point. (F) Immunohistochemistry staining for Iba-1 (violet) and orexin-A (brown) in the hypothalamus of control and Orex-HA mice at Day 14. Scale bar = 50 µm. (G) Representative histograms of MHC class II expression on myeloid cell subsets in the brain of control and Orex-HA mice 2 weeks after Pandemrix® immunization. (H) Proportion of CD45^{dim} CD11b⁺ microglia expressing MHC class II molecules in control and Orex-HA mice, 14 days after Pandemrix® immunization. Box plots are depicting interquartile range and median, and whiskers extend between min and max values of 8 to 12 mice per group from three independent experiments. Statistical analyses were performed using the Mann-Whitney test. ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001.

neurons would occur in vaccinated Orex-HA mice. We, therefore, quantified orexin-producing neurons in the lateral hypothalamus of Orex-HA recipient mice 2 months after Pandemrix® vaccination. Strikingly, an average 44% loss of orexinergic neurons was observed in the Orex-HA recipients as compared to control mice, whereas the neighbouring MCH-producing neurons were spared from destruction, suggesting an antigen-specific killing process (Fig. 3F). No significant differences in quantity of vigilance states and sleep architecture were found between vaccinated Orex-HA and control mice. Although no patent clinical manifestations were detected, polysomnographic analyses combined to video recordings revealed a cataplectic episode in two out of five vaccinated Orex-HA mice, but none in the four vaccinated control animals (Fig. 3G).

To evaluate whether the vaccine adjuvant, AS03, could also induce hypothalamic inflammation, Orex-HA mice and wild-type littermates received both HA-specific CD4 and CD8 T cells and were injected the next day with PBS, the AS03 adjuvant or the Pandemrix® vaccine. The full vaccine induced the phenotype described above, whereas no T-cell infiltration developed in the hypothalami of PBS-injected Orex-HA recipient mice (Supplementary Fig. 3C). Immunization

with the AS03 adjuvant alone elicited inconstant and very modest hypothalamic T-cell infiltration in Orex-HA mice (Supplementary Fig. 3C), mostly located in the meninges, associated with limited microglial activation (Supplementary Fig. 3D and E). Furthermore, no degenerating Orexin+ neurons were detected in these mice (*n* = 3 mice per group at Day 28).

Together, our data indicate that the HA-specific CD4 and CD8 T cells activated by the full Pandemrix® vaccine elicited hypothalamic inflammation only in mice expressing HA as a neo self-antigen. This antigen-dependent and focal inflammation resulted in the selective demise of orexinergic neurons, akin to human NT1.

Molecular characterization of hypothalamus-infiltrating T cells in vaccinated Orex-HA mice

To identify the T cell subsets and molecular pathways potentially involved in the killing of orexinergic neurons in Pandemrix®-vaccinated Orex-HA mice, we cell-sorted activated (CD44^{high} CD62L^{low}) transferred (CD45.1+) and endogenous (CD45.1-) CD4 and CD8

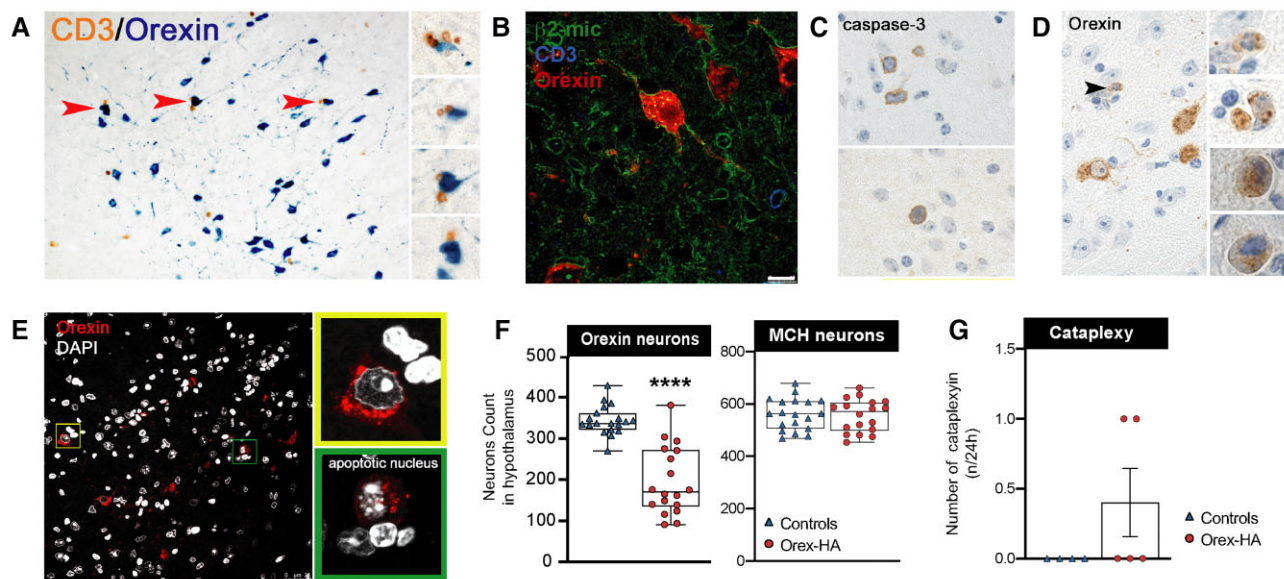


Figure 3 Hypothalamic inflammation induced by Pandemrix[®] vaccination of Orex-HA mice results in the selective loss of orexin-producing neurons. (A) Immunohistochemistry staining of CD3 (brown) and Orexin (blue) showing close contacts between T cells and orexinergic neurons (red arrow). Right insets are depiction of T cells in close apposition with orexinergic neurons in the hypothalamus of Orex-HA recipients. (B) Immunofluorescence staining for Orexin (red), β 2-microglobulin (green) and CD8 (blue) shows the presence of MHC-class I+ orexinergic neurons in the inflamed hypothalamus. (C) Images showing neurons with activated caspase-3, indicative of apoptosis. (D) Illustration of Orexin neurons showing orexin+ neurons with weak labelling for orexin (arrow head) and the presence of apoptotic nuclei as seen in the four photographs at higher magnification. (E) Immunofluorescence staining for Orexin (red) and nuclei (DAPI: white). Yellow inset (top) shows a normal orexinergic neuron while the green inset (bottom) shows a degenerating neuron with weak labelling for orexin and the presence of apoptotic bodies in the nucleus. (A–E) Photographs were taken in hypothalamus of Orex-HA animals 2 weeks after transfer of both HA-specific CD4 and CD8 T cells and Pandemrix[®] immunization. (F) Count of orexin- and MCH-producing neurons within the hypothalamus of control (blue triangles) and Orex-HA mice (red circles) 60 days after HA-specific CD4 and CD8 T-cell transfer and Pandemrix[®] immunization. Box plots are depicting interquartile range and median, and whiskers extend between min and max values of 18 to 19 mice per group from four independent experiments. Statistical analyses were performed using the Mann-Whitney test. **** $P < 0.0001$. (G) Enumeration of cataplectic events in four control (blue triangles) and five Orex-HA mice (red circles) by polysomnographic analysis combined to video-monitoring.

T cells from dissected hypothalami, spleen and cervical lymph nodes at the peak of hypothalamic infiltration (Day 14 post-vaccination) (Supplementary Fig. 4A). RNA sequencing on these sorted T-cell populations allowed us to compare the gene expression profile of a given T-cell subset across different tissue sites and of different T-cell subsets in a given tissue.

The principal component analysis (PCA) showed that, while CD8 or CD4 T cells clustered separately as expected, the major variance was found when comparing hypothalamic T cells to their peripheral counterparts, and, intriguingly, hypothalamic endogenous and transferred CD8 or CD4 T cells exhibited closely related transcriptional signature (Fig. 4A). Indeed, a cluster dendrogram highlighted the transcriptomic proximity between hypothalamus-infiltrating CD8 T cells of both endogenous and transferred origin (Fig. 4B). We identified 1247 DEG when comparing transferred CD8 T cells from the hypothalamus to those from the spleen. A similar analysis yielded 1333 DEG for the endogenous CD8 T cells (Supporting Data S1). Over representation analysis of the DEG highlighted KEGG pathways involved in immune responses and inflammatory diseases, most notably cytokine/cytokine receptor interaction (Fig. 4C). As illustrated in the heat map (Fig. 4D), transferred HA-specific CD8 T cells in the hypothalamus were characterized by an activated phenotype (*Il2ra*, *Ctla4*, *Pdcd1*, *Tigit*, *CD38*, *Irf8*, *Fos*, *Jun*, etc), with tissue-migration properties (*Sell*, *Jam2*, *Mmp2*) and an inflammatory and cytotoxic potential (*Ifng*, *CCL3*, *CCL4*, *Csf2*, *Gzmb*, *FasL*).

The hypothalamus-infiltrating endogenous CD8 T cells and the transferred CD8 T cells shared 345 DEG when compared to their splenic counterparts. Gene-set enrichment analyses

revealed that the transcriptomic signature of transferred CD8 T cells in the hypothalamus, relative to their peripheral counterparts, was largely shared by hypothalamic endogenous CD8 T cells (Fig. 4E). This similarity was also apparent when analysing the expression of selected genes (Fig. 4D). These data suggest that both endogenous and transferred CD8 T cells harbor a pathogenic potential.

The quantitatively modest transferred CD4 T cells found within the hypothalamus was also characterized by DEG involved in the highlighted KEGG pathways related to immune responses and inflammatory or infectious diseases (Supplementary Fig. 4B). Interestingly, this hypothalamic CD4 T cell subset expressed increased levels of cytotoxic and inflammatory transcripts (*Tnf*, *Tnfsf10*, *FasL*, *Gzmb*, *Ifng*, *Csf2*, *CXCL10*, *CCL3*, *CCL4*), but also transcripts related to immune regulation and tissue repair (*IL-10*, *IL-10ra*, *Areg*) as compared to CD4 T cells residing in secondary lymphoid organs (Supplementary Fig. 4C).

Respective role of HA-specific CD4 and CD8 T cells in the pathogenic process

Since the RNAseq analyses showed that hypothalamus-infiltrating CD8 and CD4 T cells expressed genes involved in effector functions, we assessed which T-cell subsets were necessary to elicit hypothalamic inflammation. Orex-HA mice were transferred with either HA-specific CD4 or CD8 T cells, or not transferred prior to Pandemrix[®] vaccination. The endogenous T-cell repertoire of Orex-HA mice was insufficient *per se* to induce focal hypothalamic

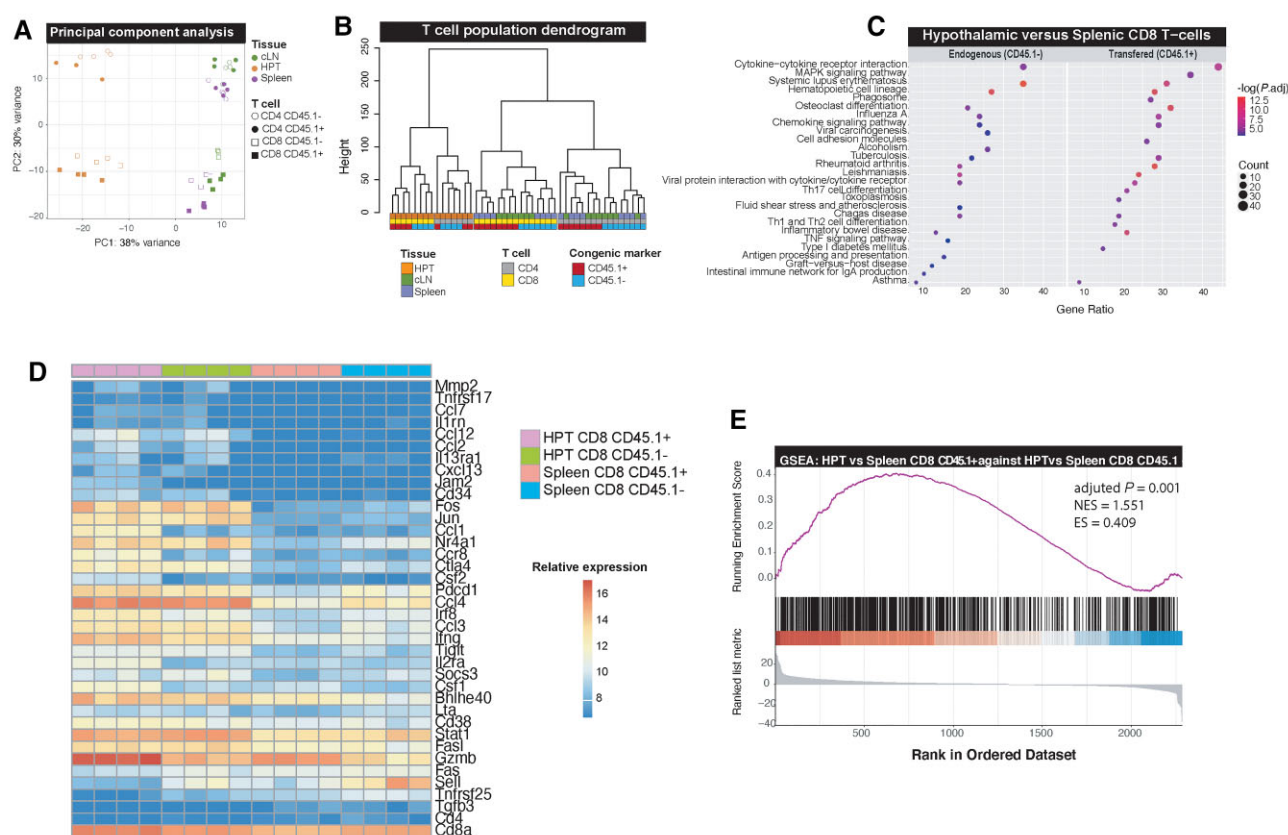


Figure 4 Analysis of the gene expression profile of hypothalamus-infiltrating T cells compared to that of their peripheral counterparts. (A) PCA of the samples originating from hypothalamus (Hypo), cervical lymph nodes (cLN) and spleen. PC1 displays 38% variance that can be explained by the separation of the three tissues. PC2 shows 30% variance that can be explained by the separation between the CD4 and CD8 T-cell subsets. (B) Dendrogram of all samples displaying the gene expression proximity of the samples according to T-cell subtype and location. Height has been calculated as the mean Euclidean distance between two sets of leaves. (C) Top 20 over-representation analysis KEGG pathways generated from the DEGs (Padj. cut-off <0.05 and fold change >1.5) between hypothalamus versus spleen endogenous CD8 T cells (left) or transferred CD8 T cells (right). (D) Heat map of genes of interest chosen from the top KEGG pathways. (E) GSEA plot showing the enrichment of the DEGs from hypothalamus versus spleen transferred CD8 T cells within the DEGs from hypothalamus versus spleen endogenous CD8 T cells. ES = enrichment score; NES = normalized enrichment score.

inflammation after vaccination (Fig. 5A). In Orex-HA mice supplemented only with HA-specific CD8 T cells, the transferred CD8 T cells initially underwent an intense proliferation and differentiation, similar to that observed in the presence of transferred CD4 T cells (Supplementary Fig. 5A–D). However, the subsequent functions of these ‘unhelped’ HA-specific CD8 T cells were altered as they failed to induce hypothalamic T-cell infiltration in Orex-HA mice after Pandemrix® vaccination (Fig. 5A). Consequently, the density of orexin+ neurons remained in the normal range in Orex-HA mice, not transferred with T cells or transferred only with CD8 T cells prior to Pandemrix® vaccination (Fig. 5C). This contrasted with the data shown above in which co-transfer of HA-specific CD8 T cells with HA-specific CD4 T cells resulted in consistent hypothalamic T-cell infiltration (Fig. 2B), including 32% of transferred HA-specific CD8 T cells (Fig. 2E), leading to significant loss of orexin+ neurons (Fig. 3F).

Interestingly, Orex-HA mice supplemented only with HA-specific CD4 T cells developed hypothalamic T-cell infiltration of varying magnitude that, however, remained significantly less than what was detected in mice supplemented with both HA-specific CD4 and CD8 T cells (Figs 5A and 2B). Endogenous CD8 T cells dominated this infiltration (Fig. 5B). Importantly, the density of orexin+ neurons in Orex-HA mice receiving HA-specific CD4 T cells

was decreased by 33% on average, whereas MCH+ neurons were spared (Fig. 5C).

These data indicate that transferred HA-specific CD4 T cells are necessary for T-cell infiltration in the hypothalamus of vaccinated Orex-HA mice and for the subsequent selective destruction of orexinergic neurons. The transferred HA-specific CD4 T cells could be either directly pathogenic or act indirectly through another immune cell subset. To address this question, given that the endogenous hypothalamic CD8 T cells exhibited a transcriptomic profile similar to that of ‘pathogenic’ transgenic HA-specific CD8 T cells, we further assessed whether depletion of CD8 T cells would prevent the destruction of orexinergic neurons induced by the Pandemrix® vaccination in Orex-HA mice supplemented with HA-specific CD4 T cells. Antibody-mediated depletion of CD8 T cells was started 5 days after Pandemrix® vaccination, prior to parenchymal T-cell infiltration of the hypothalamus. This treatment resulted in a clear reduction in the number of CD8 T cells infiltrating the hypothalamus, with no reduction of endogenous and transferred CD4 T cells (Supplementary Fig. 5E and F). Importantly, the reduction of orexinergic neuron density induced by transferred HA-specific CD4 T cells and vaccination was significantly prevented by antibody-mediated CD8 T-cell depletion (Fig. 5D). These data indicate that CD8 T cells are necessary in mediating neuronal damage and that

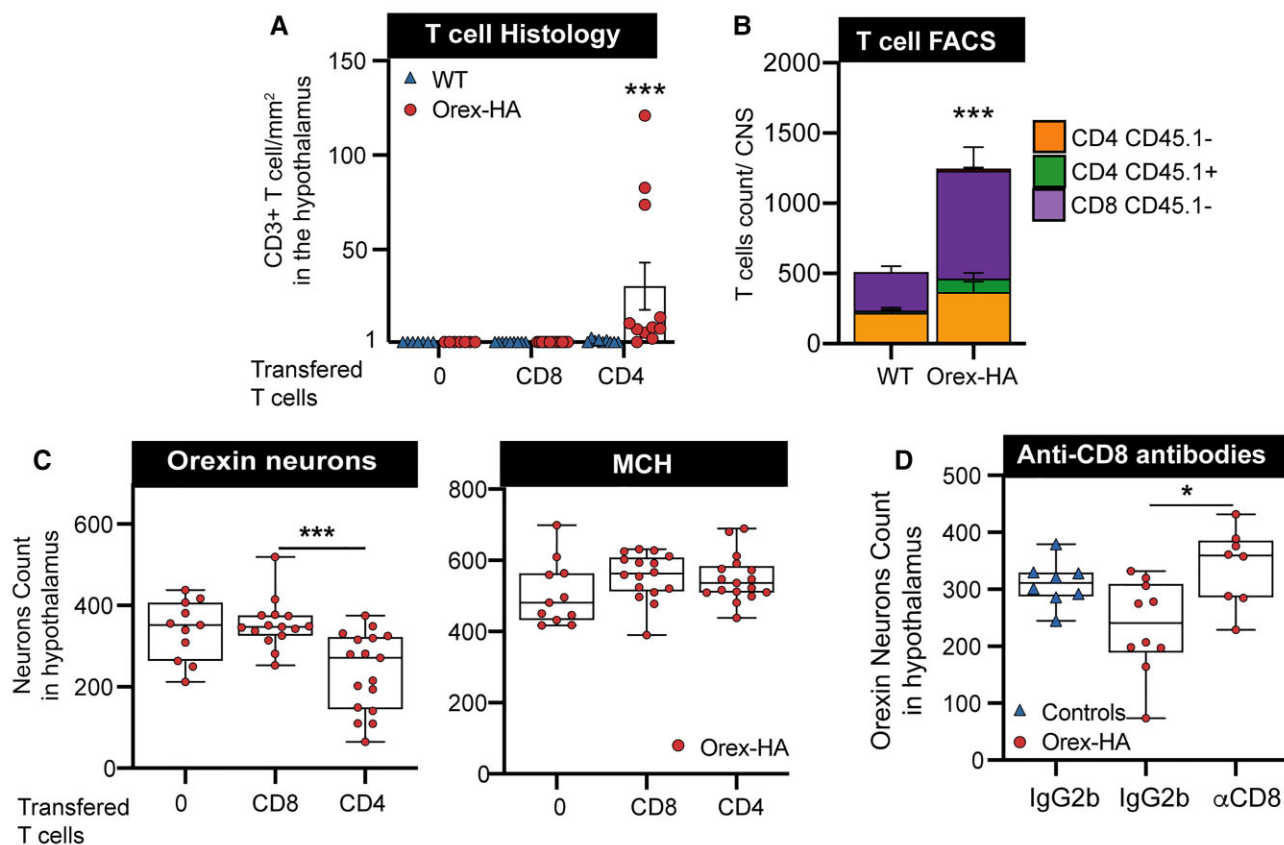


Figure 5 The loss of orexinergic neurons induced by transferred CD4 T cells upon Pandemrix® immunization requires either transferred or endogenous CD8 T cells. (A) CD3 T cell density in the hypothalamus of control (blue triangles) and Orex-HA (red circles) mice transferred with no T cells (0), HA-specific CD8 T cells, or HA-specific CD4 T cells, 14 days after Pandemrix® immunization. Results are expressed as mean $\pm$ SEM of 7 to 13 mice per group from four independent experiments. (B) Numbers of infiltrating endogenous and transgenic T cells within the whole CNS of control or Orex-HA mice transferred with HA-specific CD4 T cells only, 14 days after Pandemrix® immunization. Results are expressed as mean $\pm$ SEM of 8 to 10 mice per group from two independent experiments. (C) Count of orexin- and MCH-producing neurons within the hypothalamus of Orex-HA mice (red circles) 60 days after transfer of either no T cells (0), HA-specific CD8 T cells or HA-specific CD4 T cells and Pandemrix® immunization. Box plots are depicting interquartile range and median, while whiskers extend between min and max values of 11 to 17 mice per group. (D) Orexin neuron count within the hypothalamus of control (blue triangle) or Orex-HA mice (red circles) 4 weeks after transfer of HA-specific CD4 T cells, Pandemrix® immunization and treatment with either a depleting anti-CD8 antibody or control IgG2b from Day 5 onwards. Results are from two independent experiments. Statistical analyses were performed using the Mann–Whitney test. * $P < 0.05$, *** $P < 0.001$.

the endogenous CD8 T-cell population can substitute for the transgenic HA-specific CD8 T cells. Collectively, the data suggest a step-wise scenario in which transferred HA-specific CD4 T cells are necessary for the entry of CD8 T cells in the hypothalamus of vaccinated Orex-HA mice, whereas CD8 T cells act as the final actors of orexinergic neuronal loss.

Essential role for IFN γ produced by CD8 T cells in the pathogenic process

We characterized further the endogenous T-cell populations infiltrating the hypothalamus of vaccinated Orex-HA mice supplemented with both HA-specific CD4 and CD8 T cells. Based on TCR gene segment usage and rearrangement extracted from the RNAseq datasets, the endogenous CD4 and CD8 T-cell compartments appear diverse (Supplementary Fig. 6A–D). The diversity of the endogenous CD4 T-cell compartment was further illustrated by the presence of sizeable proportion of Foxp3-expressing regulatory cells (Supplementary Fig. 6E), as well as a significant subset responding to HA peptide stimulation by production of pro-inflammatory cytokines such as IFN γ , TNF α and GM-CSF (Supplementary Fig. 6F).

We then investigated the activation status and the mechanisms whereby CD8 T cells destroy the orexin+ neurons. In line with the transcriptomic data (Fig. 4D), both transferred and endogenous CD8 T cells produced higher levels of granzyme B in the hypothalamus than in the spleen (Fig. 6A and B). Of note, *in situ* granzyme B polarization from CD8 T cells towards orexin-producing neurons was detected in vaccinated Orex-HA mice (Fig. 6C), suggesting a direct cytotoxic effect of the tissue-infiltrating CD8 T cells following TCR:MHC-peptide interaction. Moreover, a majority of transferred and endogenous CD8 T cells, and a large proportion of CD4 T cells, from the hypothalamus produced high levels of inflammatory cytokines upon polyclonal activation (Supplementary Fig. 7A and B). A sizable fraction of endogenous (on average 29%) and transferred (on average 11.4%) CD8 T cells expressed the cell proliferation-associated Ki67 marker, suggesting *in situ* antigen stimulation. This led us to evaluate, by fluorescence activated cell sorting, whether hypothalamus-infiltrating CD8 T cells responded to HA antigen (Fig. 6D). As expected, a strong antigen-specific response was elicited in transferred (CD45.1⁺) CD8 T cells (Fig. 6E). Notably, a distinct subset of endogenous (CD45.1⁺) CD8 T cells was able to produce IFN γ and TNF α upon *ex vivo* stimulation with an

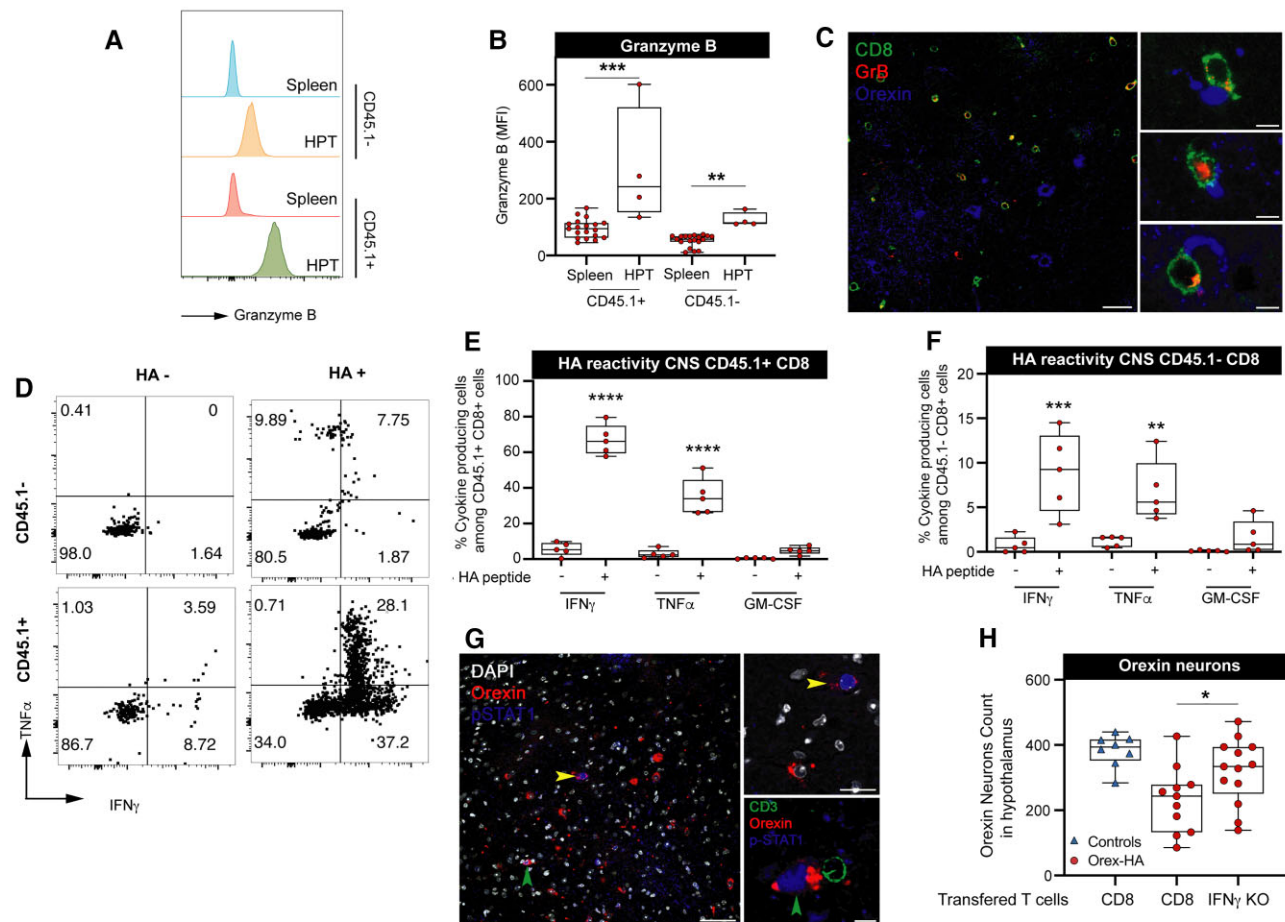


Figure 6 CD8 T cells act as final effectors of orexinergic neuron destruction through an IFN γ -dependent pathway. (A) Representative histogram and (B) quantification of granzyme B expression in transferred (CD45.1+) and endogenous (CD45.1-) CD8 T cells from the hypothalamus and spleen of Orex-HA mice 14 days after transfer of both HA-specific CD4 and CD8 T cell and Pandemrix[®] immunization. Box plots are depicting interquartile range and median, while whiskers extend between min and max values. Individual data points are plotted for spleen ($n = 19$ mice) while pools of four to eight mice were used for hypothalamus. (C) Confocal fluorescence staining of CD8 (green), Granzyme B (red) and Orexin (blue) shows the distribution of cytotoxic T cells in the lateral hypothalamus of Orex-HA mice 14 days after HA-specific T-cell transfer and Pandemrix[®] immunization. Scale bar = 25 μ m. The three enlargements on the right show GrB+ CD8 T cells attached to Orexin+ neurons or processes. Scale bars = 5 μ m. (D–F) Representative fluorescence activated cell sorting plot (D) and quantification of production of IFN γ , TNF α and GM-CSF upon HA stimulation of transferred CD45.1+ (E) and endogenous CD45.1- (F) CD8 T cells from the hypothalamus of Orex-HA mice 28 days after transfer of HA-specific T cell and Pandemrix[®] immunization. Box plots depict interquartile range and median, while whiskers extend between minimum and maximum values of five mice per group from two independent experiments. (G) Overview of the orexinergic nucleus stained for Orexin (red), p-STAT-1 (blue) and nuclear counterstain DAPI (white). Neurons indicated by the yellow and green arrowhead show upregulation of p-STAT-1. Scale bar = 50 μ m. The top inset shows an enlargement of the neuron indicated by yellow arrowhead. Scale bar = 20 μ m. The bottom inset shows the p-STAT-1+ orexin+ neuron, indicated by the green arrowhead, combined with a staining for CD3+ T cell. Scale bar = 10 μ m. (H) Orexin neurons count 60 days after transfer of HA-specific CD4 T cells in combination with either IFN γ -deficient or IFN γ -proficient HA-specific CD8 T cells. Box plots depict interquartile range and median, whiskers extend between minimum and maximum values of 8 to 13 mice per group from two independent experiments. Statistical analyses were performed using the Mann-Whitney U-test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

immunodominant HA peptide (Fig. 6F). The identification of orexinergic neurons expressing a phosphorylated form of STAT1 (an indirect marker of IFN γ receptor activation) in vaccinated Orex-HA mice suggested IFN γ signalling within neurons (Fig. 6G). Since neurons can express MHC class I molecules upon IFN γ stimulation, we wondered whether the strong IFN γ production by hypothalamus-infiltrating CD8 T cells could fuel the pathogenic process. We, therefore, transferred Orex-HA mice with HA-specific CD4 T cells and either wild-type or IFN γ -deficient HA-specific CD8 T cells prior to Pandemrix[®] vaccination. Although hypothalamic T-cell infiltration was similar in both groups at day 30 (IFN γ -proficient CD8 T cells, $n = 4$, $m = 17.67 \pm 5.15/\text{mm}^2$ vs. IFN γ -KO CD8 T cells, $n = 6$, $m = 17.75 \pm 5.75/\text{mm}^2$;

$P = 0.99$), a significant reduction in the loss of orexin-producing neurons was observed at day 60 (Fig. 6H). These data indicate that CD8 T cell-derived IFN γ plays an important role in neuronal loss in the animal model.

Discussion

In this work, we illuminate the immune mechanisms potentially linking Pandemrix[®] vaccination to hypothalamic inflammation and the eventual loss of orexin-producing neurons, a pathognomonic feature of NT1. Given the overwhelming genetic evidence indicating a role of T cells in human narcolepsy, we focused on this cell type using a dedicated animal model. Pandemrix[®] vaccination

of Orex-HA mice enriched in naïve HA-specific CD4 and CD8 T cells resulted in hypothalamic inflammation and selective destruction of orexin⁺ neurons. Transcriptomic analysis of both transferred and endogenous CD4 and CD8 T cells from hypothalamus or secondary lymphoid organs highlighted the remarkable singularity of hypothalamus-infiltrating endogenous and transferred T cells. The contribution of each T cell subset in the development of this pathogenic process was investigated using transfer of single populations of T cells as well as antibody-mediated depletion of given T-cell populations. Transferred HA-specific CD4 T cells were necessary for the development of hypothalamic inflammation but required CD8 T cells, either endogenous or adoptively transferred, to carry out the destruction of orexinergic neurons. The propensity of hypothalamic CD8 T cells to produce IFN γ in response to HA peptide stimulation and the indication that IFN γ signalling was operating in orexinergic neurons pointed to a role for IFN γ released by T cells in pathogenesis. This work demonstrates that an immunopathological process mimicking narcolepsy can be elicited by the Pandemrix[®] vaccine in an experimental mouse model. Our work also highlights the synergy between autoreactive CD4 and CD8 T cells for disease development: CD4 T cells being essential for the development of hypothalamic inflammation and CD8 T cells being responsible for the final execution of orexin-producing neurons.

Molecular mimicry is usually suspected when an autoimmune process develops following a documented infection or vaccination.²⁶ This mechanism has been confirmed for a number of neurological autoimmune diseases, such as Guillain-Barré syndrome occurring after *Campylobacter jejuni* infection, in which antibodies against bacteria lipo-oligosaccharides cross-react with peripheral nerve gangliosides²⁷ or Sydenham's chorea related to a cross-reactivity between Group A streptococcal epitopes and neuronal antigens.²⁸ Related to this mechanism, paraneoplastic diseases can develop as a result of *de novo* expression of a (neuronal) autoantigen by tumoral cells leading to activation of pathogenic autoreactive T cells.²⁹ More broadly, instances in which self-reactive T-cell clones from individuals with autoimmune diseases are shown to cross-react with multiple foreign or even self-antigens are now becoming uncountable.^{30,31} Molecularly, this observation is related to the overall promiscuity of MHC-peptide recognition by a given TCR. However, to our knowledge, there are no proven instances of pathogenic cross-reactivity between a vaccine viral antigen and a human self-antigen. The search for molecular mimicry in post-Pandemrix[®] narcolepsy has been active. A striking study reported detection of autoantibodies against the orexin receptor type 2 in 85% of post-Pandemrix[®] NT1 paediatric patients in Finland (*n* = 20), which cross-reacted with the flu vaccine nucleoprotein.³² Similar data were not replicated in other patient cohorts.³³ Interestingly, although cross-reactivity was not functionally demonstrated, TCR sequences of CD4 T cell clones recognizing C-amidated forms of orexin peptides in association with HLA-DQ6 shared striking similarities with those from influenza HLA-DQ6:HA-specific clones.¹⁹ These data speak in favour of a molecular mimicry mechanism underlying the development of narcolepsy in persons vaccinated with Pandemrix[®] during the 2009/2010 pandemic. Yet they do not prove causality. Our data go one step further by showing that orexinergic neuron destruction may result from molecular mimicry between the vaccine HA antigen and a similar/identical antigen expressed primarily in orexinergic neurons. Nevertheless, we did not test in our mouse model whether exposure to the infectious H1N1 'influenza' virus could elicit, or contribute to, a similar phenotype.^{34,35}

Our previous study showed that adoptively transferred autoreactive Th1 cells could not promote orexin⁺ neuron death despite their presence in the hypothalamus, whereas transfer of cytotoxic CD8 T cells could induce both inflammation and death of orexin⁺ neurons.¹⁶ The current work suggests a subtler role for autoreactive CD4 T cells, elicited by the flu vaccine. Indeed, CD4 T cells were essential to endow the autoreactive CD8 T cells with the capacity to enter the hypothalamus. In their absence, transferred CD8 T cells proliferated strongly but were harmless, presumably because the endogenous CD4 T cell help induced by the vaccine was limiting. Although these 'unhelped' transferred CD8 T cells acquired an antigen-experienced phenotype and could produce IFN γ , their functional properties were likely altered as they failed to penetrate and/or remain in the hypothalamic parenchyma. These data are in line with previous data underlining the importance of CD4 T cell help in conferring tissue invasiveness of CD8 T cells recognizing the same immunizing antigen, through up-regulation of chemokine receptors and matrix metalloproteases.³⁶ In addition, following vaccination in absence of CD4 T cell help, vaccine-specific CD8 T cells exhibited decreased cytotoxic potential and expressed higher levels of co-inhibitory receptors such as PD-1, LAG-3, Tim-3, 2B4 and BTLA.^{36–38} CD4 T cell help is mostly provided through activation of dendritic cells that relay CD28 and CD27-mediated signals in CD8 T cells.³⁹ In our mouse model, both transferred and endogenous CD8 T cells expressed high levels of granzyme B, FasL, IFN γ and CCL3, CCL4 in the hypothalamus. This is indicative of a pro-inflammatory and cytotoxic potential. Moreover, evidence for granzyme-mediated cytotoxicity of the targeted neurons was provided by neuropathological analyses. Then, once in the hypothalamus, converging evidence indicate that the cytotoxic CD8 T cells mediate antigen-specific killing of orexinergic neurons. The concept that antigen-specific T cells can directly target CNS neurons has received strong support from experimental models of CNS infection or autoimmunity.^{40–43} Moreover, accumulating data strongly suggest that a similar mechanism could be at play in human inflammatory or degenerative neurological diseases, including NT1 but also amyotrophic lateral sclerosis, Parkinson's disease and Alzheimer's disease.^{42,44–49} One unexpected result was the observation that the transferred CD4 T cells could also promote neuronal cell death, in the absence of transferred HA-specific CD8 T cells, by licensing the endogenous CD8 T cells to kill. Indeed, the endogenous CD8 T-cell compartment contains HA-specific T cells likely elicited by the Pandemrix[®] vaccine; these cells, as shown by the CD8 T cell-depletion experiments, were necessary for orexinergic neuron loss. The contribution of the endogenous CD4 T-cell population infiltrating the hypothalamus to the disease process is less clear. Indeed, this subset is composed of at least three different populations: (i) HA-specific CD4 T cells that could very well contribute to the inflammation and destruction of orexinergic neurons; (ii) bystander activated CD4 T cells of unrelated specificity that have the propensity to release IFN γ and TNF within the hypothalamus and could, thereby, contribute to the pathogenic process as suggested in the context of experimental autoimmune encephalomyelitis⁵⁰; and (iii) regulatory CD4 T cells expressing Foxp3 that are recruited locally.

Our results are in line with the strong association of NT1 with the HLA-DQA1*01:02-HLA-DQB1*06:02 haplotype and with the increased frequency of orexin-specific CD4 T cells found in the blood of NT1 patients versus healthy donors.^{18,19} Indeed, the increased number of autoreactive CD4 T cells can contribute to break the mechanisms of peripheral immune tolerance thereby enabling CD8 T cells to migrate within the hypothalamus.

T cells can release IFN γ directionally, promoting molecular interactions with cells presenting the recognized MHC-peptide ligands.⁵¹ This event could well occur in the hypothalamus of our mouse model, as orexinergic neurons exhibit nuclear localization of phosphorylated STAT1, an indirect indication of IFN γ -R triggering. Other investigators have observed expression of pSTAT1 in neurons in experimental models of CD8 T cell-mediated experimental CNS autoimmune diseases as well as in human diseases associated with prominent CD8 T cell brain inflammation.^{40,43} As a result of IFN γ signalling, CNS neurons can engage several pro-inflammatory pathways that will culminate in disruption of neuronal connectivity and neuronal death. First, neurons can release chemokines, such as CCL2 and CXCL10, which will attract myeloid cells as well as additional T cells in their vicinity. Second, IFN γ signalling promotes MHC class I expression by neurons, hence promoting their recognition by antigen-specific cytotoxic CD8 T cells. Finally, autocrine or paracrine action of IFN γ enhances *in vivo* mobility of cytotoxic lymphocytes and promotes target killing.⁵² The importance of this pathway is underlined in our model by the preventive impact of IFN γ deficiency on the loss of neurons but not on CD8 T cell penetration in the hypothalamus. The importance of the IFN γ pathway in pathogenesis is reinforced by data from patients with NT1. Indeed, serum levels of IFN γ were higher in NT1 patients, in particular those with recent onset, as compared to healthy controls,⁵³ although this was not detected in CSF.⁵⁴ Intriguingly, the serum levels of CXCL9 and CXCL10, two IFN γ -induced chemokines were higher in post-H1N1 vaccine NT1 as compared to non-exposed NT1 patients.⁵³ Moreover, higher frequencies of orexin-specific T cells able to produce IFN γ were detected in patients with NT1 as compared to healthy controls.^{18,55} Based on these observations, we surmise that IFN γ neutralization could represent an attractive therapeutic strategy for neurological autoimmune diseases including, but not restricted to, NT1. However, this therapeutic approach would have to be tested at the onset of disease before major and irreversible destruction of neuronal populations occur.

Acknowledgements

We thank Drs D. Gonzalez-Dunia and A. Dejean for their insightful comments on the manuscript. We are indebted to the Cytometry platform of Infinity and to Anne-Laure Morel and Rebecca Boston from CRNL for their technical help. We thank the UMS06 for mouse care and histological platform.

Funding

Pandemrix® is a trademark of the GSK group of companies. This work was supported by Inserm, CNRS, and grants from Agence Nationale pour la Recherche (CE14-14066; CE17-0014), Toulouse University, Fondation Bettencourt-Schueller (Emergence Idex), ERA-Net Narcomics and GlaxoSmithKline. The funders had no role in the design and interpretation of the study.

Competing interests

This study was partially funded by GlaxoSmithKline.

Supplementary material

Supplementary material is available at *Brain* online.

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